Transformer Network: 30 Questions & Answers

1. What is tokenization in a transformer network?

Tokenization is the process of converting text into smaller units called tokens, which can be words, subwords, or characters, for processing by the model.

2. What is Byte Pair Encoding (BPE)?

BPE is a tokenization method that merges the most frequent pairs of bytes in a dataset, allowing efficient handling of rare words and subword units.

3. Are tokenization parameters trainable?

No, tokenization itself has no trainable parameters; it is a preprocessing step.

4. What is an embedding in the context of transformers?

An embedding is a learned vector representation of a token, capturing its semantic meaning.

5. How are similar words represented in embedding space?

Similar words (like "dog" and "cat") have similar embedding vectors, meaning they are close in the embedding space.

6. What is positional embedding?

Positional embedding is a vector added to token embeddings to encode the position of each token in the sequence.

7. Why are positional embeddings needed?

Transformers lack recurrence or convolution, so positional embeddings allow the model to understand the order of tokens.

8. Are positional embeddings trainable?

Yes, positional embeddings are learned parameters during training.

9. What is the attention mechanism?

Attention allows each token to focus on (or "attend to") other tokens in the sequence when forming its representation.

10. What is the key innovation of the transformer architecture?

The transformer replaces recurrence with attention, allowing parallel computation and better long-range dependency modeling.

11. What is an attention score?

An attention score quantifies the relevance between two tokens, computed as a function of their representations.

12. How are attention scores used?

Attention scores are used to weigh and aggregate information from other tokens when updating a token’s representation.

13. What is an attention matrix?

An attention matrix contains the attention scores between all pairs of tokens in a sequence.

14. What is multi-head attention?

Multi-head attention computes attention scores in parallel using different sets of parameters, capturing diverse types of relationships.

15. Why use multi-head attention?

It allows the model to focus on different aspects of the sequence simultaneously, improving expressiveness.

16. What is a transformer block?

A transformer block is a stackable module consisting of multi-head attention, feed-forward layers, and normalization.

17. What is the feed-forward layer in transformers?

It is a position-wise fully connected layer applied to each token's representation after attention.

18. How are transformer blocks stacked?

Multiple transformer blocks are stacked to build deep models capable of complex reasoning.

19. What is the output layer of a transformer?

Typically, it is a linear layer followed by a softmax for language modeling or classification tasks.

20. How does a transformer aggregate information?

By computing weighted sums of token representations using attention scores.

21. What is the input to a transformer model?

A sequence of token embeddings plus positional embeddings.

22. What is the main advantage of transformers over RNNs?

Transformers allow parallel processing of all tokens, leading to faster training and better long-range dependency modeling.

23. What is the computational complexity of self-attention?

Self-attention has O(n^2) complexity with respect to sequence length n.

24. What is the role of softmax in attention?

Softmax normalizes attention scores to sum to 1, forming a probability distribution.

25. How does the transformer handle variable-length sequences?

By masking padding tokens and only attending to real tokens.

26. What is layer normalization?

A normalization technique applied within transformer blocks to stabilize training.

27. What is the difference between encoder and decoder in transformers?

The encoder processes the input sequence, while the decoder generates the output sequence, attending to both input and previous outputs.

28. Give an example of a transformer-based model.

BERT, GPT, and T5 are all transformer-based models.

29. What is the function of the lookup table in embedding layers?

It maps token indices to their corresponding embedding vectors.

30. Why is context important in transformers?

Context allows each token's representation to be influenced by other relevant tokens in the sequence.

Naïve Bayes Classifier: 30 Questions & Answers

1. What is the main idea behind Naïve Bayes classifiers?

To estimate the probability of a class given features using Bayes' theorem, assuming feature independence.

2. What is Bayes' Rule?

P(Y|X) = P(X|Y) * P(Y) / P(X)

3. What does "naïve" refer to in Naïve Bayes?

The assumption that all features are conditionally independent given the class label.

4. What is the Maximum A Posteriori (MAP) estimate?

Choosing the class with the highest posterior probability given the features.

5. How is the class prior probability estimated?

$P(Y=y)$ = Number of records with class y / Total number of records

6. How is $P(X_i|Y=y)$ estimated for discrete attributes?

By counting: $P(X_i=x|Y=y)$ = Number of records with $X_i=x$ and $Y=y$ / Number of records with $Y=y$

7. How is $P(X_i|Y=y)$ estimated for continuous attributes?

Assume a distribution (e.g., normal), estimate its parameters (mean, variance), and compute the probability density.

8. What is Laplace smoothing?

A technique to avoid zero probabilities by adding 1 to counts: P = (count + 1) / (total + number of possible values)

9. Why is Laplace smoothing necessary?

To handle unseen attribute values in the test data that were not present in the training data.

10. What is conditional independence?

Two features are conditionally independent given the class if $P(X1|X2, Y) = P(X1|Y)$.

11. What is the formula for the joint probability under the independence assumption?

$P(X1, X2, ..., Xd|Y) = P(X1|Y) * P(X2|Y) * ... * P(Xd|Y)$

12. How is the class of a new instance predicted?

By computing $P(Y|X)$ for each class and choosing the one with the highest value.

13. What is the effect of the independence assumption on model accuracy?

It simplifies computation but may reduce accuracy if features are actually correlated.

14. How are categorical features handled?

By counting occurrences for each value-class combination.

15. How are continuous features handled?

By fitting a probability distribution (typically normal) and using its density function.

16. What is the mean and variance formula for MLE?

Mean: $\mu = (1/n) \Sigma x_i$; Variance: $\sigma^2 = (1/n) \Sigma (x_i - \mu)^2$

17. Give an example of a Naïve Bayes application.

Spam detection, where words are features and the class is spam/not spam.

18. What is the main computational advantage of Naïve Bayes?

Fast training and prediction due to independence assumption.

19. How does Naïve Bayes handle missing data?

By ignoring missing features when computing probabilities.

20. What is the output of a Naïve Bayes classifier?

Class probabilities for each possible class.

21. What is the effect of highly correlated features?

It can degrade model performance due to the violated independence assumption.

22. How does Naïve Bayes handle multi-class classification?

By computing P(Y=class|X) for each class and picking the highest.

23. What is the role of the denominator in Bayes' rule?

It normalizes the probabilities so they sum to 1.

24. What is the difference between prior and posterior probability?

Prior is before observing data; posterior is after observing data.

25. What is the computational complexity of Naïve Bayes?

Linear in the number of features and data points.

26. What is the zero-frequency problem?

When a feature value never occurs with a class in training, leading to zero probability.

27. How does Naïve Bayes perform with small datasets?

It can perform well, especially with Laplace smoothing.

28. Can Naïve Bayes be used for regression?

No, it is a classification algorithm.

29. What is the effect of redundant features?

Redundant features can bias the probability estimates.

30. How do you interpret the output probabilities?

They represent the model's confidence in each class given the input features.

K-means Clustering: 30 Questions & Answers

1. What is clustering?

Grouping objects such that objects in the same group are similar, and objects in different groups are dissimilar.

2. What is the main objective of k-means clustering?

To minimize the sum of distances from each point to its assigned centroid.

3. What is a centroid?

The mean position of all points assigned to a cluster.

4. What is the first step in k-means?

Randomly select K points as initial centroids.

5. How are points assigned to clusters?

Each point is assigned to the closest centroid.

6. What happens after assignment in k-means?

Centroids are recomputed as the mean of assigned points.

7. When does k-means stop iterating?

When centroids no longer change (convergence).

8. What is the time complexity of k-means?

O(nki*d), where n = number of points, k = clusters, i = iterations, d = dimensions.

9. How is the number of clusters (K) chosen?

It must be specified in advance, often using methods like the elbow method.

10. What is the "elbow method"?

A technique to choose K by plotting the sum of squared errors and looking for a point where the decrease slows.

11. What is intra-cluster distance?

The distance between points within the same cluster.

12. What is inter-cluster distance?

The distance between points in different clusters.

13. What is the main limitation of k-means?

It assumes clusters are spherical and equally sized.

14. How does k-means handle outliers?

Poorly; outliers can distort centroids.

15. What is the effect of initialization in k-means?

Different initial centroids can lead to different results (local minima).

16. What is k-means++?

An improved initialization method that spreads out initial centroids.

17. Can k-means handle categorical data?

No, it is designed for numerical data.

18. How is the distance measured in k-means?

Typically using Euclidean distance.

19. What happens if K is too small?

Clusters may be too broad and not capture the structure.

20. What happens if K is too large?

Clusters may be too specific, capturing noise.

21. Can k-means find non-convex clusters?

No, it works best for convex, spherical clusters.

22. What is the objective function in k-means?

The sum of squared distances from each point to its centroid.

23. What is a cluster assignment step?

Assigning each point to the nearest centroid.

24. What is a centroid update step?

Recomputing centroids as the mean of assigned points.

25. What is a convergence criterion in k-means?

Centroids do not change between iterations.

26. How does k-means scale with data size?

It is efficient for large datasets but can be slow for very high dimensions.

27. What is the role of random seeds in k-means?

They control the initial centroid selection for reproducibility.

28. Can k-means be used for image compression?

Yes, by clustering pixel colors.

29. What is the effect of duplicate points in k-means?

They can bias centroid locations.

30. How do you evaluate k-means clustering quality?

Using metrics like within-cluster sum of squares or silhouette score.

Boosting: 30 Questions & Answers

1. What is boosting?

An ensemble technique that combines multiple weak learners to form a strong learner.

2. What is a weak learner?

A model that performs slightly better than random guessing.

3. What is the main idea of boosting?

To sequentially train weak learners, each focusing on the mistakes of the previous ones.

4. What is AdaBoost?

A popular boosting algorithm that adjusts weights on training examples.

5. How are sample weights updated in AdaBoost?

Weights increase for misclassified examples and decrease for correctly classified ones.

6. What is the final prediction in boosting?

A weighted vote (or sum) of the predictions from all weak learners.

7. What is the main advantage of boosting?

It can significantly improve model accuracy.

8. What is overfitting in boosting?

Boosting can overfit if too many weak learners are added or if they are too complex.

9. What is gradient boosting?

A boosting technique where each new learner fits the residual errors of the previous model.

10. What is XGBoost?

An efficient and scalable implementation of gradient boosting.

11. What is the loss function in boosting?

Typically, exponential loss (AdaBoost) or other differentiable losses (gradient boosting).

12. How does boosting handle noisy data?

It can overfit to noise, so regularization is important.

13. What is bagging vs boosting?

Bagging trains models in parallel on random samples; boosting trains sequentially, focusing on errors.

14. What is the base learner in boosting?

Usually a decision stump or shallow tree.

15. How does boosting improve bias and variance?

It reduces bias and can also reduce variance compared to a single weak learner.

16. What is early stopping in boosting?

Halting training when validation error stops improving to prevent overfitting.

17. What is the effect of too many weak learners?

Potential overfitting and increased computation.

18. Can boosting be used for regression?

Yes, with modifications to the loss and aggregation functions.

19. How does boosting handle class imbalance?

By increasing the weights of misclassified minority class examples.

20. What is the difference between AdaBoost and gradient boosting?

AdaBoost reweights samples; gradient boosting fits residuals.

21. What is the role of the learning rate in boosting?

It controls the contribution of each weak learner.

22. What is shrinkage in boosting?

Multiplying each weak learner's prediction by a small learning rate to improve generalization.

23. How do you prevent overfitting in boosting?

Use early stopping, regularization, and limit the depth of weak learners.

24. What is the bias-variance tradeoff in boosting?

Boosting reduces bias but can increase variance if overfitting occurs.

25. What is the effect of noisy features in boosting?

Boosting can overfit to noise, so feature selection is important.

26. What is the computational cost of boosting?

Higher than a single model due to sequential training.

27. Can boosting be parallelized?

Not easily, as each learner depends on the previous one.

28. What is the output of a boosting model?

A weighted sum or vote of all weak learners' predictions.

29. How is feature importance measured in boosting?

By the frequency and impact of features used in weak learners.

30. Name a real-world application of boosting.

Fraud detection, where boosting is used to combine multiple simple rules.

Model Generalization: 30 Questions & Answers

1. What is generalization in machine learning?

The ability of a model to perform well on unseen data.

2. What is overfitting?

When a model performs well on training data but poorly on new data.

3. What is underfitting?

When a model is too simple to capture the underlying patterns in the data.

4. What is the bias-variance tradeoff?

The balance between a model's ability to fit training data (bias) and its sensitivity to fluctuations (variance).

5. What is regularization?

A technique to penalize model complexity to improve generalization.

6. What is L2 regularization?

Adding the squared magnitude of coefficients as a penalty to the loss function.

7. What is L1 regularization?

Adding the absolute value of coefficients as a penalty to the loss function.

8. What is dropout?

A regularization technique where random neurons are ignored during training.

9. What is early stopping?

Halting training when validation performance stops improving.

10. What is cross-validation?

A method to estimate model performance by splitting data into training and validation sets multiple times.

11. What is a validation set?

A subset of data used to tune model parameters and estimate generalization.

12. What is a test set?

A subset of data used only to assess the final model's performance.

13. What is data augmentation?

Increasing training data by creating modified versions of existing data.

14. What is model capacity?

The ability of a model to fit a wide variety of functions.

15. What is the effect of high capacity?

Higher risk of overfitting.

16. What is the effect of low capacity?

Higher risk of underfitting.

17. What is cross-entropy loss?

A loss function often used for classification tasks.

18. What is the role of hyperparameter tuning in generalization?

Selecting optimal model parameters to balance bias and variance.

19. What is ensembling?

Combining multiple models to improve generalization.

20. What is bagging?

An ensemble method that trains models on random subsets of data.

21. What is the effect of noisy labels on generalization?

It can degrade model performance.

22. What is data leakage?

When information from outside the training dataset is used to create the model, leading to over-optimistic performance.

23. What is the difference between training and validation error?

Training error is on seen data; validation error estimates performance on unseen data.

24. What is model selection?

Choosing the best model based on validation performance.

25. What is the purpose of a learning curve?

To visualize model performance as a function of training set size.

26. What is transfer learning?

Using a model trained on one task as the starting point for a related task.

27. What is the effect of data imbalance on generalization?

Models may perform poorly on minority classes.

28. How can you address data imbalance?

By resampling, using class weights, or data augmentation.

29. What is the role of feature selection in generalization?

Removing irrelevant features can improve generalization.

30. What is the effect of model complexity on generalization?

Too complex: overfitting; too simple: underfitting.

Evaluation Metrics: 30 Questions & Answers

1. What is accuracy?

The proportion of correct predictions among all predictions.

2. What is precision?

The proportion of true positives among all positive predictions.

3. What is recall?

The proportion of true positives among all actual positives.

4. What is the F1 score?

The harmonic mean of precision and recall.

5. What is a confusion matrix?

A table showing the counts of true positives, false positives, true negatives, and false negatives.

6. What is specificity?

The proportion of true negatives among all actual negatives.

7. What is sensitivity?

Another term for recall.

8. What is the ROC curve?

A plot of true positive rate vs. false positive rate at various thresholds.

9. What is AUC?

Area Under the ROC Curve; a measure of overall model performance.

10. What is the PR curve?

Precision-Recall curve, useful for imbalanced datasets.

11. What is log loss?

A loss function for probabilistic classification models.

12. What is mean squared error (MSE)?

The average of squared differences between predicted and actual values.

13. What is mean absolute error (MAE)?

The average of absolute differences between predicted and actual values.

14. What is R^2 score?

A measure of how well regression predictions approximate real values.

15. What is balanced accuracy?

The average of recall obtained on each class.

16. What is macro-averaging?

Computing metrics independently for each class and averaging.

17. What is micro-averaging?

Aggregating contributions of all classes to compute the average metric.

18. What is weighted averaging?

Averaging metrics weighted by the number of true instances per class.

19. What is Cohen's kappa?

A statistic measuring inter-rater agreement for categorical items.

20. What is Matthews correlation coefficient?

A balanced measure for binary classification, even with class imbalance.

21. What is the G-mean?

The geometric mean of sensitivity and specificity.

22. What is the effect of class imbalance on accuracy?

Accuracy can be misleading if one class dominates.

23. How do you choose an evaluation metric?

Based on the problem, data distribution, and business goals.

24. What is the threshold in classification?

The probability cutoff for assigning a class label.

25. What is the effect of changing the threshold?

It changes precision and recall trade-off.

26. What is the purpose of cross-validation in evaluation?

To get a reliable estimate of model performance.

27. What is overfitting in the context of evaluation metrics?

When a model scores high on training metrics but low on validation/test metrics.

28. What is a baseline metric?

A simple metric (e.g., majority class accuracy) used for comparison.

29. What is a cost-sensitive metric?

A metric that accounts for different costs of false positives/negatives.

30. What is the importance of using multiple metrics?

To get a comprehensive view of model performance.